Practical 6

Considered there are N philosophers seated around a circular table with one chopstick between each pair of philosophers. There is one chopstick between each philosopher. A philosopher may eat if he can pick up the two chopsticks adjacent to him. One chopstick may be picked up by any one of its adjacent followers but not both.

Write a program to solve the problem using process synchronization technique.

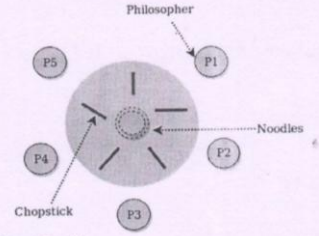

Fig: 1. Dining Philosopher Problem

CODE -

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5

sem_t chopstick[N];

void* philosopher(void* num)
{
    int id = *(int*)num;

    while(1)
    {
        printf("Philosopher %d is Thinking\n", id);
        sleep(1);

        if(id == N-1)
        {
            sem_wait(&chopstick[0]);
```

```
            sem_wait(&chopstick[id]);
        }
        else
        {
            sem_wait(&chopstick[id]);
            sem_wait(&chopstick[id+1]);
        }

        printf("Philosopher %d is Eating\n", id);
        sleep(2);

        if(id == N-1)
        {
            sem_post(&chopstick[0]);
            sem_post(&chopstick[id]);
        }
        else
        {
            sem_post(&chopstick[id]);
            sem_post(&chopstick[id+1]);
        }

        printf("Philosopher %d finished Eating\n", id);
    }
}

int main()
{
    pthread_t thread[N];
    int id[N];

    for(int i=0; i<N; i++)
        sem_init(&chopstick[i], 0, 1);
```

```
for(int i=0; i<N; i++)
{
    id[i] = i;
    pthread_create(&thread[i], NULL, philosopher, &id[i]);
}

for(int i=0; i<N; i++)
    pthread_join(thread[i], NULL);

return 0;
```

Output –

```
Philosopher 0 is Thinking
Philosopher 1 is Thinking
Philosopher 2 is Thinking
Philosopher 3 is Thinking
Philosopher 4 is Thinking
Philosopher 0 is Eating
Philosopher 3 is Eating
Philosopher 3 finished Eating
Philosopher 4 is Eating
Philosopher 3 is Thinking
Philosopher 2 is Eating
Philosopher 0 finished Eating
Philosopher 0 is Thinking
Philosopher 4 finished Eating
Philosopher 4 is Thinking
Philosopher 2 finished Eating
Philosopher 2 is Thinking
Philosopher 1 is Eating
Philosopher 3 is Eating
Philosopher 1 finished Eating
Philosopher 1 is Thinking
Philosopher 0 is Eating
Philosopher 3 finished Eating
Philosopher 3 is Thinking
Philosopher 2 is Eating
Philosopher 0 finished Eating
```

```
Philosopher 0 is Thinking
Philosopher 4 is Eating
Philosopher 1 is Eating
Philosopher 2 finished Eating
Philosopher 2 is Thinking
Philosopher 4 finished Eating
Philosopher 3 is Eating
Philosopher 0 is Eating
Philosopher 4 is Thinking
Philosopher 1 finished Eating
Philosopher 1 is Thinking
Philosopher 3 finished Eating
Philosopher 3 is Thinking
Philosopher 0 finished Eating
Philosopher 0 is Thinking
Philosopher 2 is Eating
Philosopher 4 is Eating
Philosopher 2 finished Eating
Philosopher 2 is Thinking
Philosopher 4 finished Eating
Philosopher 4 is Thinking
Philosopher 3 is Eating
Philosopher 1 is Eating
Philosopher 3 finished Eating
Philosopher 3 is Thinking
Philosopher 1 finished Eating
```

```
Philosopher 1 is Thinking
Philosopher 0 is Eating
Philosopher 2 is Eating
Philosopher 0 finished Eating
Philosopher 1 is Eating
Philosopher 4 is Eating
Philosopher 0 is Thinking
Philosopher 2 finished Eating
Philosopher 2 is Thinking
Philosopher 1 finished Eating
Philosopher 1 is Thinking
Philosopher 4 finished Eating
Philosopher 4 is Thinking
Philosopher 0 is Eating
Philosopher 3 is Eating
Philosopher 3 finished Eating
Philosopher 4 is Eating
Philosopher 2 is Eating
Philosopher 0 finished Eating
Philosopher 0 is Thinking
Philosopher 3 is Thinking
Philosopher 4 finished Eating
Philosopher 4 is Thinking
Philosopher 2 finished Eating
Philosopher 2 is Thinking
Philosopher 3 is Eating
```

```
Philosopher 1 is Eating
Philosopher 3 finished Eating
Philosopher 3 is Thinking
Philosopher 1 finished Eating
Philosopher 1 is Thinking
Philosopher 0 is Eating
Philosopher 2 is Eating
Philosopher 2 finished Eating
Philosopher 3 is Eating
Philosopher 0 finished Eating
Philosopher 0 is Thinking
Philosopher 2 is Thinking
Philosopher 1 is Eating
Philosopher 3 finished Eating
Philosopher 4 is Eating
Philosopher 3 is Thinking
Philosopher 1 finished Eating
Philosopher 1 is Thinking
Philosopher 2 is Eating
Philosopher 4 finished Eating
Philosopher 4 is Thinking
Philosopher 2 finished Eating
Philosopher 2 is Thinking
Philosopher 3 is Eating
Philosopher 1 is Eating
Philosopher 3 finished Eating
```